Roadmap To India's SEMICONDUCTOR PRODUCTISATION

As India strides towards semiconductor prowess, it invites us to ponder: Can this journey redefine global innovation and economic landscapes?

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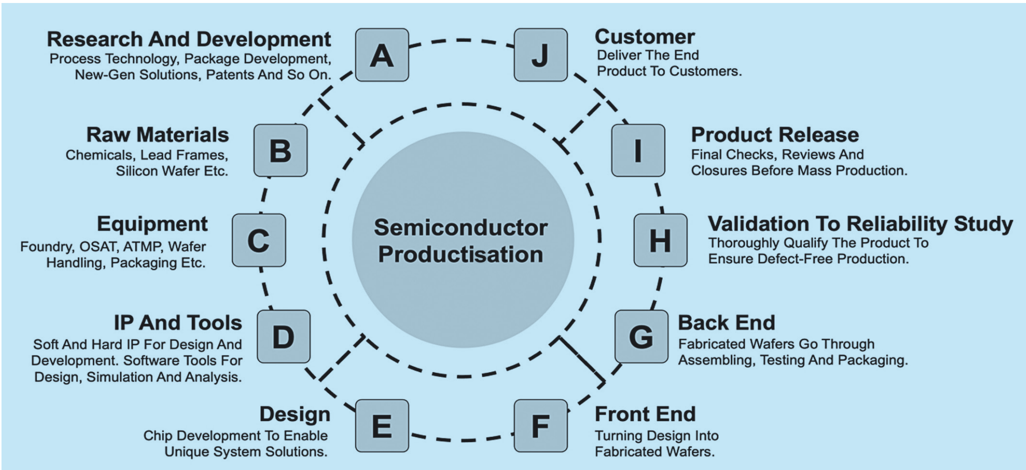

India, traditionally known for its prowess in software services, is now ambitiously moving towards becoming a significant player in the global semiconductor industry. This industry, which is expected to reach a market size of over $600 billion globally by 2026, represents a golden opportunity for India to diversify and elevate its role in the high-tech sector

Semiconductor ambition of India

India's foray into the semiconductor industry is timely and strategically imperative. In an era where semiconductors are the backbone of virtually all modern technologies—from smartphones, wireless communication, and electric vehicles to AI—the demand for these components is surging globally, making nations more dependent on them. As the world increasingly contends with supply chain disruptions and geopolitical tensions impacting semiconductor production and distribution, India's entry into this sector could bolster its technological sovereignty and position it as a crucial player in the global supply chain.

Moreover, India's established strengths in the software sector provide a robust foundation for growth in the semiconductor domain, including the vast pool of engineering talent that accelerates its journey from being a consumer to a creator of semiconductor technology. This transition from software to hardware and services to manufacturing could mark a significant turning point in India's technological and economic trajectory.

However, a critical aspect India should address before becoming a semiconductor product nation is semiconductor productisation. India has been home to top semiconductor design centres, including startups catering well to semiconductor product development's design aspect. However, building a semiconductor product involves multiple steps. Ultimately, it entails transforming any silicon idea into a manufacturable product, which includes several critical stages of development to ensure the design is defect-free when mass-produced. In the long run, a nation that can complete all these steps for 100% of the productisation process in-country will lead to semiconductor technological advancement.

What is semiconductor productisation

Semiconductor productisation is critical in the silicon technology value chain because it transforms theoretical and technical capabilities into tangible semiconductor products. This process requires synergy between engineering, manufacturing, marketing,